# TARUNKUMAR PALANIVELAN

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## EDUCATION

**New York University - Tandon School of Engineering, New York** Pursuing
*Master of Science, Mechatronics and Robotics* (2024-2026)
**SRM Institute of Science and Technology, Chennai** 2020-2024
*Bachelor of Technology, Mechatronics with Specialization in Robotics* GPA: 8.43/10

## EXPERIENCE

***Graduate Research Assistant* - Agile Robotics and Perception Lab, New York University** Sept 2024 - Present
- Designed mounts and vibration-isolated assemblies for cameras, IMUs, and computer systems, for the UAV research platform.
- Performed sensor integration of depth cameras, IMUs for VIO-based autonomous flights and state estimation.
- Assisted in experiments, executing pre-flight checks, calibration, and avionics validation(IMU alignment, motor tests, GPS lock).
- Supported EKF-based localization**,** validating multi-sensor fusion pipelines and alignment of visual-inertial sensors.

***Teaching Assistant* - Mechatronics , New York University** Sept 2025 - Present
- Guided students through sensor integration, microcontroller programming, and embedded system debugging, including IMUs, encoders, sensors, motors, and communication protocols (UART, I2C, SPI).
- Assisted students in implementing EKF sensor fusion for localization using IMUs, depth cameras, and wheel odometry.
- Evaluated student designs, ensuring wiring practices, grounding, noise reduction, and microcontroller-to-sensor interfacing.
- Led troubleshooting sessions focused on circuit safety, power systems, microcontroller programming, and modelling of a system.

***Robotics & Perception Engineer* - NYU Self-Drive Team** Sept 2025 – Present
- Integrated LiDAR, RealSense depth cameras, IMUs, and wheel odometry on TurtleBot and FrodoBot robots.
- Built ROS2 perception stacks: LiDAR scan matching, depth processing, EKF fusion, obstacle avoidance, and path planning.
- Developed RTAB-Map-based 3D mapping for Earth Rover Challenge autonomous navigation.

***Manufacturing and Assembly Intern* - Garuda Aerospace, Chennai** June 2023 - July 2023
- Integrated avionics and mechanical components on Medium-class drones, including GPS, telemetry, and LiPo power systems.
- Conducted assembly, inspection, wiring, soldering, and structure validation for UAV hardware.
- Assisted in mission planning and pre-flight checks for mapping and surveillance UAVs.

***Team Lead* - Aerospace Systems Research Lab (SRM)** May 2021 – May 2024
- Led the mechanical design of multiple UAV and UGV platforms, producing CAD assemblies, 2D drawings and manufacturing documentation for carbon-fiber frames, motor mounts, landing gear, and sensor brackets.
- Directed electronics–mechanical integration including Pixhawk Cube, GPS modules, FPV cameras, ESCs, telemetry, and vibration-isolated IMU assemblies, ensuring proper packaging, airflow, wiring routing, and mechanical stability.
- Supervised a 40-member cross-functional team (mechanical, electrical, embedded) and coordinated design reviews, safety checks, component fabrication, and mission-readiness validation.

## TECHNICAL SKILLS

**Mechanical Design:** SolidWorks | Fusion 360 | GD&T | tolerance analysis | 2D drawings | topology optimization | static FEA.
**Autopilot & Sensors:** Pixhawk Cube, Here3 GPS, RGBD cameras, 2D LiDAR, IMUs, wheel encoders, actuators.
**Embedded & Compute:** ROS2 | PX4 | ArduPilot | MAVLink | Mission Planner | QGC | EKF/UKF sensor fusion.
**Software & Programming:** Python | MATLAB | C/C++ | Linux | Rviz | Gazebo | NumPy/Pandas | data logging & analysis.

**CERTIFICATIONS** - FAA TRUST | CSWA Mechanical | CSWA Additive | CSWA Electrical | NPTEL IoT
**ACHIEVEMENTS** - 1st Place – Rotorcraft 2024 | Top 5 (World) – Technoxian | Top 6 India – IEEE Yesist12

## PROJECTS

**Parachute Fail-Safe System for UAVs.** *Jan 2024 – Apr 2024*
- Designed a real-time parachute deployment fail-safe system for drones using a Propeller microcontroller, external IMU, and the UAV's onboard flight controller IMU for redundant safety sensing, implemented Mars-Rover–inspired descent sequencing, altitude-triggered deployment and state transitions.
- Integrated a distance sensor to measure ground proximity and trigger a servo-actuated retractable landing gear for softer landings.
- Developed firmware for fall detection, motor-power-loss detection, emergency override, and safe system disarm conditions.

**Hybrid Reconfigurable UAV–UGV (Morphing Robot)**
- Designed complete 3D CAD assemblies of a morphing aerial–ground robot with servo-actuated thrust-tilting mechanisms.
- Performed structural FEA on parts, including motor mounts, servo housings, and landing gear, validating factors of safety and deformation; Conducted topology optimization to reduce weight by 50% while maintaining structural rigidity.
- Performed CG estimation, balancing UAV and UGV configurations for stability and evaluated multiple iterations to optimize weight, stiffness, and manufacturability; Executed validation tests, logging thrust, current, voltage sag, and operational stability

**ZEUS – High-Performance Hexacopter (Rotorcraft 2024 Winner)**
- Designed the full mechanical structure of a custom hexacopter, achieving an 1:1 payload-to-drone weight ratio, enabling ZEUS to carry a 900 g payload on a 900–1000 g airframe, a key factor in winning Best Design at Rotorcraft 2024.
- Conducted topology optimization on structural plates and landing gear; achieved lightweight geometry while maintaining rigidity under combined 120 N motor thrust loads; Performed complete CG estimation using component weight breakdown and datum offsets; refined electronics placement for balanced stability.
- Executed autonomous mission testing validating hover power, endurance, thrust margins, and performing a precision payload drop mission for competition scoring.